

Homework 2

APPM 4490/5490 Theory of Machine Learning, Spring 2026

Due date: Wednesday, Jan 28 '26, before noon, via paper or via Gradescope

Instructor: Prof. Becker

Revision date: 1/21/2026

Theme: Uniform convergence and measure theory

Instructions Collaboration with your fellow students is OK and in fact recommended, although direct copying is not allowed. The internet is allowed for basic tasks (e.g., looking up definitions on wikipedia) but it is not permissible to search for proofs or to *post* requests for help on forums such as <http://math.stackexchange.com/> or to look at solution manuals. Please write down the names of the students that you worked with.

An arbitrary subset of these questions will be graded.

Reading You are responsible for chapters 1, 2, 3 and up to 4.1 in “Understanding Machine Learning” by Shai Shalev-Shwartz and Shai Ben-David (2014, Cambridge University Press). Note: you can buy the book for about \$45 on Amazon (or less for an e-book), and the authors host a free PDF copy on their [website](#) (but note that this PDF has different page numbers).

Problem 1: Read the paper “[Uniform convergence may be unable to explain generalization in deep learning](#)” by Vaishnavh Nagarajan and J. Zico Kolter, which was awarded the “Outstanding new directions paper” at NeurIPS 2019. Please read this paper (you do *not* have to read the appendix), and turn in a response to this paper with 2 to 3 thoughtful paragraphs. *Use of any AI for this assignment is not allowed.*

Some possible prompts you can respond to if you need ideas:

- They prove some technical results, but their focus is on the high-level result that 2-sided uniform convergence cannot adequately explain observations. They argue that their technical result implies this high-level summary, but this argument is subtle. Do you buy that they have proved their high-level argument?
- The first paragraph of Section 2 discusses the bounds that depend on norm-based quantities. Does the authors’ argument depend on these bounds being “tight”? (and what does “tight” mean?)
- Do you think the large margin “gamma” in section 2 could explain any of the results? Is this a standard way to train?
- Does Figure 1 need error bars? Or is it believable as is? Do you think behavior might change once m gets *really* large?
- Why do they analyze a surrogate loss (especially if $\gamma = 0$ covers the usual 0-1 loss)?
- Footnote 2: they pursue another possible explanation. Does this mean there could be more explanations they didn’t pursue?
- The example for Theorem 3.1 requires training for exactly one epoch? Does that have any effect? Also, rather than use noisy labels for their examples (like other papers), they use “extraneous” or “distracting” features. Is this a fair setup? Also, they have the class centers have a norm that depends on m ; is this valid?
- What does “ h ” mean in this paper? (i.e., it is overloaded: what are the two meanings?)

- i) This is a NeurIPS paper from 2019. The NeurIPS deadline is usually in May, but this preprint on arXiv was posted in February (arxiv.org/abs/1902.04742). Why do you think that is? Do you think the authors got feedback (informal, or by being rejected to a conference such as ICML, which had a Jan 18 2019 deadline that year) which shaped how they wrote the paper? Do you think this paper is more thorough than the average paper, because they are anticipating blowback?
- j) Some of the NeurIPS reviews are public. What do you think about the quality of the reviews? Does this make you confident in the accuracy of the peer review process?
<https://proceedings.neurips.cc/paper/2019/file/05e97c207235d63ceb1db43c60db7bbb-Reviews.html>.

Problem 2: Read/skim lectures 7 and 8 (but refer to part 1 (lectures 1–6) as needed) in Joel Tropp’s lecture notes [Probability Theory & Computational Mathematics, CMS/ACM 117 / Caltech / Fall 2023](#)¹. Make sure to use the version from Fall 2023 so that numbering is correct; do not use the 2021 version. We’re going to do part of Exercise 8.9, which is about proving Proposition 8.8. For a random variable X (using Tropp’s notation, which is standard probability notation; this is *not* consistent with Shalev-Shwartz and Ben-David’s notation), define the distribution function $F_X(a) = \mathbb{P}[X \leq a] = \mu_X(-\infty, a]$ as in Definition 8.7. The distribution is “right continuous,” and your job is to prove it. Specifically:

[APPM 5490 students only] Prove that if $(a_k) \subset \mathbb{R}$ is a monotonically decreasing sequence that converges to a , then

$$\lim_{k \rightarrow \infty} F_X(a_k) = F_X(a).$$

We sometimes abbreviate this to $\lim_{x \downarrow a} F_X(x) = F_X(a)$.

Hint: Read Tropp’s lecture 2, in particular section 2.3.2

[All students] Let μ be the measure of your choice on $\mathbb{R}$ (e.g., Lebesgue), and let $A_i \subset \mathbb{R}$, $i \in \mathbb{N}$, be any sequence of measurable sets. Find examples (of a measure and of a sequence of sets) such that

- a) $\mu(\bigcup_i A_i) \neq \lim_{i \rightarrow \infty} \mu(A_i)$
- b) $\mu(\bigcap_i A_i) \neq \lim_{i \rightarrow \infty} \mu(A_i)$

¹This is a hyperlink to a URL, but you may not be able to click on it in some web-based PDF viewers; in that case, download the PDF then use your regular PDF viewer.